

DAY 1 (Variables, Data Types, I/O, Operators)

1. Write a program to input two numbers and print their sum.

```
1.py  U X
Day9&10 > 1.py > ...
1  num1 = int(input("Enter first number:"))
2  num2 = int(input("Enter second number:"))
3  sum = num1 + num2
4
5  print("The sum of two numberes is:", sum)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/1.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/1.py"
Enter first number:12
Enter second number:20
The sum of two numberes is: 32
>_ ~
```

2. Write a program to calculate area of a rectangle.

```
2.py  U X
Day9&10 > 2.py > ...
1  length = float(input("Enter Length of the rectangle:"))
2  width = float(input("Enter Width of the rectangle:"))
3  area = width * length
4
5  print("Area of the rectangle is:", area)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/2.py"
Enter Length of the rectangle:12
Enter Width of the rectangle:14
Area of the rectangle is: 168.0
_
```

3. Write a program to calculate simple interest.

```
3.py  U X
Day9&10 > 3.py > ...
1  principal = float(input("Enter Principal:"))
2  rate = float(input("Enter Interest Rate:"))
3  time = float(input("Enter Duration:"))
4
5  interest = (principal * rate * time) / 100
6
7  print("Simple Interest is:", interest)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/3.py"
Enter Principal:1200
Enter Interest Rate:10
Enter Duration:1
Simple Interest is: 120.0
```

4. Write a program to convert Celsius into Fahrenheit.

```
Day9&10 > 4.py > ...
1  celsius = float(input("Enter temperature in celsius:"))
2
3  fehrenheit = ((celsius * 9) / 5) + 32
4
5  print("Temperature in fehrenheit is", fehrenheit)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/4.py"
• >_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/4.py"
Enter temperature in celsius:20
Temperature in fehrenheit is 68.0
❖ >_ ~
```

5. Write a program to input employee basic salary and calculate HRA (20%) and DA (10%).

```
5.py  U X
Day9&10 > 5.py > ...
3  hra = (salary * 20) / 100
4  da = (salary * 10) / 100
5
6  print("HRA is", hra, "DA is", da)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

• >_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/5.py"
Enter Basic salary:12000
HRA is 2400.0 DA is 1200.0
```

DAY 2 (Conditions)

6. Write a program to check whether a number is positive, negative or zero.

```
6.py U x
Day9&10 > 6.py > ...
1 num = int(input("Enter any integer:"))
2
3 if num==0:
4     print("The number is 0")
5 elif num>0:
6     print("The number is positive.")
7 else:
8     print("The number is negative.")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/6.py"
Enter any integer:12
The number is positive.
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/6.py"
Enter any integer:-14
The number is negative.
```

7. Write a program to find the largest among two numbers.

```
7.py U x
Day9&10 > 7.py > ...
1 num1 = int(input("Enter the first number:"))
2 num2 = int(input("Enter the second number:"))
3
4 if num1==num2:
5     print("Both number are same")
6 elif num1<num2:
7     print(f"{num2} is the largest number.")
8 else:
9     print(f"{num1} is the largest number.")
10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/7.py"
Enter the first number:67
Enter the second number:69
69 is the largest number.
```

8. Write a program to find the largest among three numbers.

```
8.py U x
Day9&10 > 8.py > ...
1 num1 = int(input("Enter the first number:"))
2 num2 = int(input("Enter the second number:"))
3 num3 = int(input("Enter the third number:"))
4
5 largest = max (num1, num2, num3)
6 print("Largest is", largest)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/8.py"
Enter the first number:61
Enter the second number:67
Enter the third number:69
Largest is 69
```

9. Write a program to check whether a number is even or odd.

```
9.py U X
Day9&10 > 9.py > ...
1 num = int(input("Enter an integer:"))
2
3 if num>0:
4     if num % 2 == 0:
5         print("The number is even")
6     else:
7         print("The number is odd")
8 else:
9     print("wrong input")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
> _python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/9.py"
Enter an integer:67
The number is odd
```

10. Write a program to check whether a year is a leap year or not.

```
10.py U X
Day9&10 > 10.py > ...
1 year = int(input("Enter a year:"))
2
3 if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
4     print(f"{year} is a leap year")
5 else:
6     print(f"{year} is not a leap year")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/10.py"
> _python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/10.py"
Enter a year:1999
1999 is not a leap year
> _python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/10.py"
Enter a year:2020
2020 is a leap year
```

DAY 3 (Loops)

11. Write a program to print numbers from 1 to 100.

```
11.py U 1.py X
Day9&10 > 1.py > ...
You, 4 days ago | 1 author (You)
1 num1 = int(input("Enter first number:"))
2 num2 = int(input("Enter second number:"))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/11.py"
> _python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/11.py"
1
2
3
4
5
6
```

12. Write a program to print even numbers between 1 and 100.

```
12.py U X
Day9&10 > 12.py > ...
1   for i in range(2, 101, 2):
2   |   print(i)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/12.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/12.py"
2
4
6
8
10
12
```

13. Write a program to find sum of first N natural numbers.

```
13.py U X
Day9&10 > 13.py > ...
1   n = int(input("Enter N: "))
2
3   total = 0
4   for i in range(1, n + 1):
5   |   total += i
6
7   print("Sum =", total)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/13.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/13.py"
Enter N: 12
Sum = 78
```

14. Write a program to print multiplication table of a given number.

```
14.py U X
Day9&10 > 14.py > ...
1   num = int(input("Enter a number: "))
2
3   for i in range(1, 11):
4   |   print(f"{num} x {i} = {num * i}")

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/14.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/14.py"
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

15. Write a program to calculate factorial of a number.

```
14.py U 15.py U X
Day9&10 > 15.py > ...
1   num = int(input("Enter a number: "))
2
3   factorial = 1
4
5   for i in range(1, num + 1):
6   |   factorial *= i
7
8   print("Factorial =", factorial)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/15.py"
Enter a number: 5
Factorial = 120
```

DAY 4 (Functions)

16. Write a function to find square of a number.

```
16.py U X
Day9&10 > 16.py > ...
1 def square(num):
2     return num * num
3
4 n = int(input("Enter a number: "))
5 print("Square =", square(n))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/16.py"
>_ -python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/16.py"
Enter a number: 12
Square = 144
```

17. Write a function to find maximum of two numbers.

```
17.py U X
Day9&10 > 17.py > ...
5
6 x = int(input("Enter first number: "))
7 y = int(input("Enter second number: "))
8
9 print("Maximum =", maximum(x, y))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/17.py"
>_ -python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/17.py"
Enter first number: 1
Enter second number: 2
Maximum = 2
```

18. Write a function to calculate simple interest.

```
18.py U X
Day9&10 > 18.py > ...
4 p = float(input("Principal: "))
5 r = float(input("Rate: "))
6 t = float(input("Time: "))
7
8 print("Simple Interest =", simple_interest(p, r, t))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/18.py"
>_ -python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/18.py"
Principal: 1200
Rate: 10
Time: 5
Simple Interest = 600.0
```

19. Write a function to check whether a number is even or odd.

```
19  U X
Day9&10 > 19 > ...
1  def check_even_odd(num):
2      if num % 2 == 0:
3          return "Even"
4      return "Odd"
5
6  n = int(input("Enter a number: "))
7  print(check_even_odd(n))

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/19"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/19"
Enter a number: 19
Odd _
```

20. Write a function to calculate area of a circle.

```
20.py  U X
Day9&10 > 20.py > ...
1  import math
2
3  def area_circle(radius):
4      return math.pi * radius * radius
5
6  r = float(input("Enter radius: "))
7  print("Area =", area_circle(r))

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/20.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/20.py"
Enter radius: 7
Area = 153.93804002589985
```

DAY 5 (Lists, Tuples, Dictionary, Set)

21. Write a program to create a list of 10 numbers and print all elements.

```
21.py  U X
Day9&10 > 21.py > ...
1  numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
2
3  for num in numbers:
4      print(num)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/21.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/21.py"
10
20
30
40
50
60
70
80
90
100
```

22. Write a program to find largest element in a list.

```
22.py U X
Day9&10 > 22.py > ...
1 numbers = [10, 45, 78, 23, 99, 12]
2
3 largest = max(numbers)
4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/22.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/22.py"
Largest element = 99
```

23. Write a program to calculate sum of all elements in a list.

```
23.py U X
Day9&10 > 23.py > ...
1 numbers = [10, 20, 30, 40, 50]
2
3 total = sum(numbers)
4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/23.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/23.py"
Sum = 150
```

24. Write a program to count even numbers in a list.

```
23.py U 24.py U X
Day9&10 > 24.py > ...
1 numbers = [10, 15, 20, 25, 30, 35, 40]
2
3 count = 0
4
5 for num in numbers:
6     if num % 2 == 0:
7         count += 1
8
9 print("Number of even elements =", count)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/24.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/24.py"
Number of even elements = 4
```

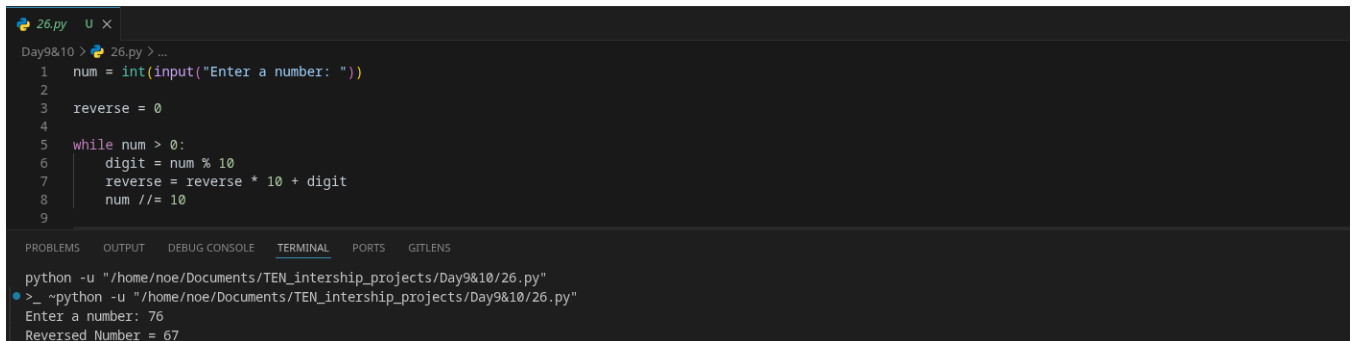
25. Write a program to remove duplicate elements using set.

```
25.py U X
Day9&10 > 25.py > ...
1 numbers = [10, 20, 30, 20, 40, 10, 50, 30]
2
3 unique_numbers = list(set(numbers))
4
5 print("Original List:", numbers)
6 print("Without Duplicates:", unique_numbers)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/25.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/25.py"
Original List: [10, 20, 30, 20, 40, 10, 50, 30]
Without Duplicates: [40, 10, 50, 20, 30]
```

10 Mixed Questions

26. Write a program to reverse a number.

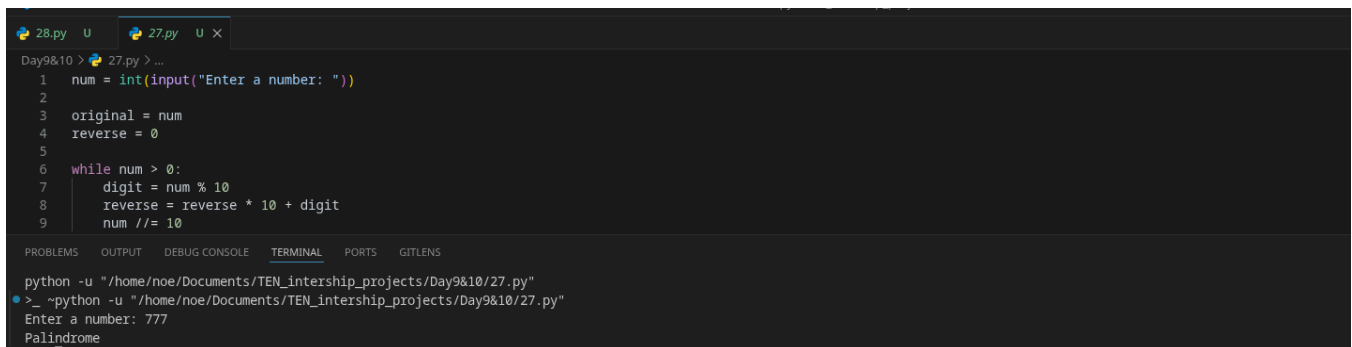


```
26.py U X
Day9&10 > 26.py > ...
1 num = int(input("Enter a number: "))
2
3 reverse = 0
4
5 while num > 0:
6     digit = num % 10
7     reverse = reverse * 10 + digit
8     num //= 10
9

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/26.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/26.py"
Enter a number: 76
Reversed Number = 67
```

27. Write a program to check whether a number is palindrome.

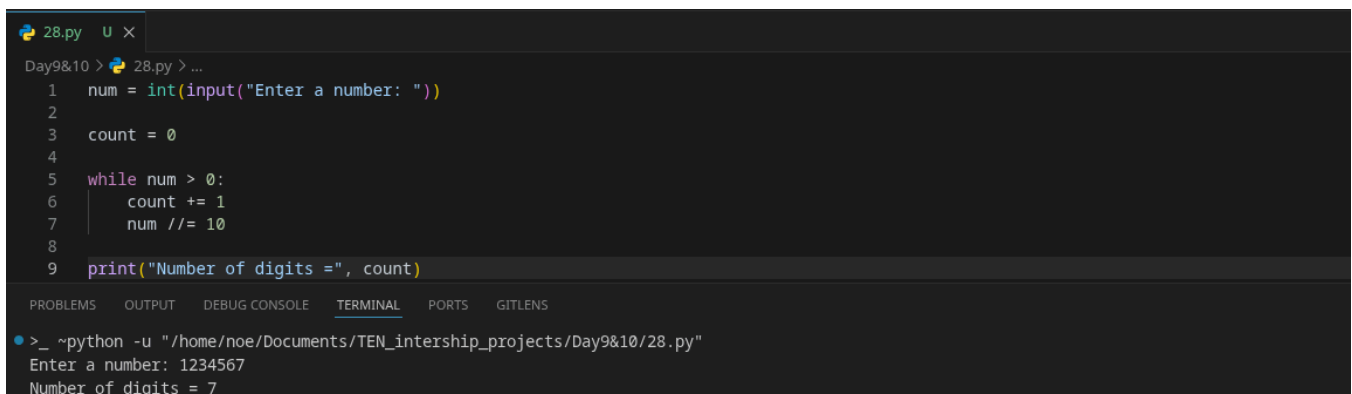


```
28.py U 27.py U X
Day9&10 > 27.py > ...
1 num = int(input("Enter a number: "))
2
3 original = num
4 reverse = 0
5
6 while num > 0:
7     digit = num % 10
8     reverse = reverse * 10 + digit
9     num //= 10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/27.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/27.py"
Enter a number: 777
Palindrome
```

28. Write a program to count digits in a number.



```
28.py U X
Day9&10 > 28.py > ...
1 num = int(input("Enter a number: "))
2
3 count = 0
4
5 while num > 0:
6     count += 1
7     num //= 10
8
9 print("Number of digits =", count)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/28.py"
Enter a number: 1234567
Number of digits = 7
```

29. Write a program to find sum of digits of a number.

```
29.py U X
Day9&10 > 29.py > ...
1 num = int(input("Enter a number: "))
2
3 total = 0
4
5 while num > 0:
6     digit = num % 10
7     total += digit
8     num //= 10
9
10 print("Sum of digits =", total)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/29.py"
> _ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/29.py"
Enter a number: 99
Sum of digits = 18
```

30. Write a program to generate Fibonacci series up to N terms.

```
30.py U X
Day9&10 > 30.py > ...
1 n = int(input("Enter number of terms: "))
2
3 a, b = 0, 1
4
5 for i in range(n):
6     print(a, end=" ")
7     a, b = b, a + b

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/30.py"
> _ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/30.py"
Enter number of terms: 30
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 >_ ~
```

31. Write a program to check whether a number is prime.

```
31.py U X
Day9&10 > 31.py > ...
1 num = int(input("Enter a number: "))
2
3 if num < 2:
4     print("Not Prime")
5 else:
6     is_prime = True
7
8     for i in range(2, int(num ** 0.5) + 1):
9         if num % i == 0:
10             is_prime = False
11             break
12
13 if is_prime:
14     print("Prime")
15 else:
16     print("Not Prime")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/31.py"
> _ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/31.py"
Enter a number: 7
Prime
```

32. Write a program to print all prime numbers

between 1 and

```
32.py U X
Day9&10 > 32.py > ...
1 for num in range(2, 101):
2     is_prime = True
3
4     for i in range(2, int(num ** 0.5) + 1):
5         if num % i == 0:
6             is_prime = False
7             break
8
9     if is_prime:
10        print(num)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/32.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/32.py"
2
3
5
7
11
```

33. Write a program to count vowels in a string.

```
33.py U X
Day9&10 > 33.py > ...
1 text = input("Enter a string: ")
2
3 count = 0
4
5 for ch in text.lower():
6     if ch in "aeiou":
7         count += 1
8
9 print("Number of vowels =", count)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/33.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/33.py"
Enter a string: TEN
Number of vowels = 1
```

34. Write a program to reverse a string.

```
34.py U X
Day9&10 > 34.py > ...
1 text = input("Enter a string: ")
2
3 reverse = text[::-1]
4
5 print("Reversed String =", reverse)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/34.py"
>_ ~python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/34.py"
Enter a string: 123
Reversed String = 321
```

35. Write a program to count frequency of each character in a string.

```
35.py U x
Day9&10 > 35.py > ...
1 text = input("Enter a string: ")
2
3 freq = {}
4
5 for ch in text:
6     freq[ch] = freq.get(ch, 0) + 1
7
8 for ch, count in freq.items():
9     print(ch, ":", count)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/35.py"
> _python -u "/home/noe/Documents/TEN_intership_projects/Day9&10/35.py"
Enter a string: TEN
T : 1
E : 1
N : 1
```